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Activity

Preparation of Solution Culture to find out Mineral Deficiency

1. Take a glass jar or polythene bottle and cover with black paper (to prevent algal growth and roots reacting with light).
2. Add nutrient solution.
3. Fix a plant with the help of split cork.
4. Fix a tube for aeration.
5. Observe the growth by adding specific minerals.

triple covalent bonds. The process of converting atmospheric nitrogen (N_2) into ammonia is termed as nitrogen fixation. Nitrogen fixation can occur by two methods: 1. Biological; 2. Non-Biological (Figure 12.5).

12.7.1 Non – Biological nitrogen fixation

- Nitrogen fixation by chemical process in industry.
- Natural electrical discharge during lightening fixes atmospheric nitrogen.

12.7.2 Biological nitrogen fixation

Symbiotic bacterium like *Rhizobium* fixes atmospheric nitrogen. Cyanobacteria found in Lichens, *Anthoceros, Azolla* and coralloid roots of *Cycas* also fix nitrogen. non-symbiotic (free living bacteria) like *Clostridium* also fix nitrogen.

a. Symbiotic nitrogen fixation

i. Nitrogen fixation with nodulation

Rhizobium bacterium is found in leguminous plants and fix atmospheric nitrogen. This kind of symbiotic association is beneficial for both the bacterium and plant. Root nodules are formed due to bacterial infection. *Rhizobium* enters into the host cell and proliferates, it remains separated from the host cytoplasm by a membrane (Figure 12.6).

Stages of Root nodule formation:

1. Legume plants secretes phenolics which attracts *Rhizobium*.
2. *Rhizobium* reaches the rhizosphere and enters into the root hair, infects the root hair and leads to curling of root hairs.
3. Infection thread grows inwards and separates the infected tissue from normal tissue.

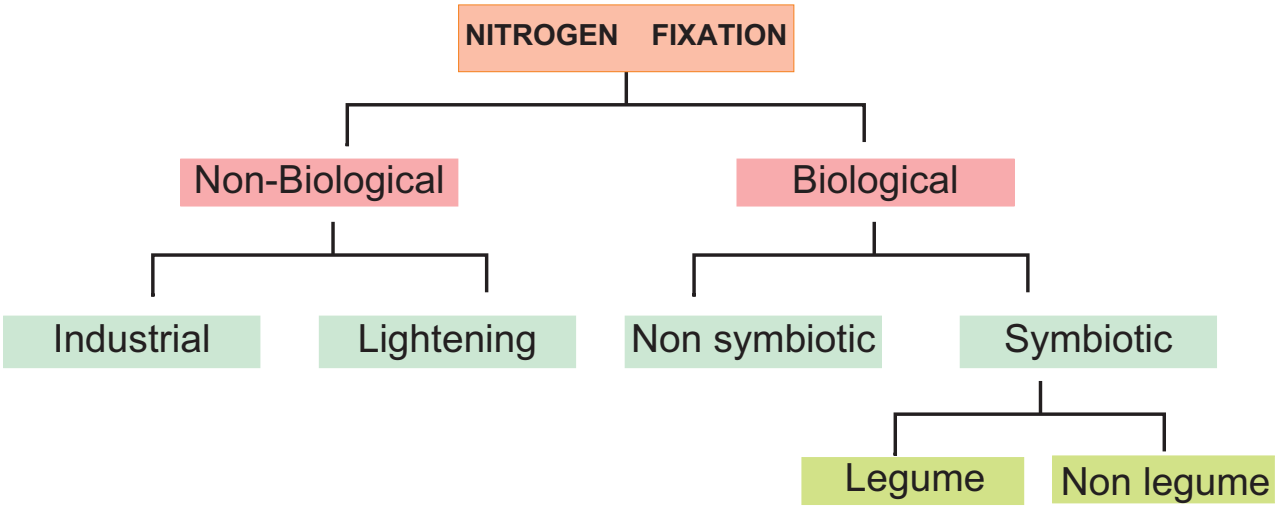


Figure 12.5: Nitrogen fixation